BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 276 · Entity relationship diagrams

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Where is a FK stored in a one-to-many relationship?

Q2. What does PK stand for?

Q3. What is an associative entity?

Q4. A BST root20 has left10 and right30; insert 25. Describe comparisons and placement.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

ERD — Entity relationship diagram.

cardinality — Number of related instances.

junction table — Table linking M:N records.

one-to-many — One entity instance can have several related rows.

New learning

Understand — ERDs model entities and relationship cardinalities before implementing tables.

Apply — One-to-many relationships are typically represented with a foreign key on the many side.

Evaluate — Many-to-many relationships require an associative/junction table when translated into a relational schema, often with two foreign keys.

Worked example

Model Author and Book when one author writes multiple books, each book has one primary author.

1. Predict first — Each Book row stores one AuthorID

2. Model answer — several books can refer to the same author.

We Do · Practise with guidance

Model a pupil taking several courses and each course having several pupils.

```
Pupil MANY ↔ MANY Course
Enrolment(PupilID FK,CourseID FK)
```

Predict first

Create Enrolment junction table

Model answer

each enrolment references exactly one pupil and course.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State meanings of 1:1,1:M and M:N cardinalities. [3]

Question 2

Draw/describe an ERD for Customer and Order with one customer placing many orders. Give foreign key. [6]

Question 3

Resolve Student many-to-many Course into relational tables; describe keys and integrity. [6]

Exit Ticket · Answer before checking

A many-to-many relationship is usually represented in relational tables by placing one simple FK on either table.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. On the many side.

Q2. Primary key.

Q3. Bridge/junction entity resolving many-to-many.

Q4. 25>20 right to30; 25<30 left child of30.

You Do pages · Independent questions

Q1. Exactly one to one [1], one related to many [1], many to many [1].

Q2. Customer 1→many Order [2]; Customer.CustomerID PK [1]; Order.OrderID PK [1]; Order.CustomerID FK [2].

Q3. Student(PupilID PK) [1], Course(CourseID PK) [1], Enrolment junction [1] with PupilID and CourseID FKs [2], referential integrity prevents orphan enrolments [1].

Exit Ticket · Model answer

False. A junction table normally represents the relationship.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: